

MATH 110C, HOMEWORK 7

DUE TUESDAY MAY 28

Exercise 1 Let K be a characteristic 0 field. Let $a, b, \varepsilon \in K$ with $\varepsilon \neq 0$.

1. Show that $K(\sqrt{a + b\sqrt{\varepsilon}})$ is a Galois extension of K with Galois group a cyclic group of order 4 if and only if ε is not a square in K and there exists $c \in K$, $c \neq 0$, such that $a^2 - \varepsilon b^2 = \varepsilon c^2$.

2. Deduce that an extension $K(\sqrt{\varepsilon})$ of K (with $\varepsilon \in K$ and ε not a square) is a subextension of a Galois extension of K with Galois group cyclic of order 4 if and only if ε is a sum of two squares in K .

Exercise 2

1. Show that the permutations (12) and $(123 \dots n)$ generate the symmetric group S_n .

2. Deduce that if p is prime and if H is a subgroup of S_p of order divisible by p and containing a transposition, then $H = S_p$.

3. Show that the Galois group of $X^5 - 4X + 2$ over $\mathbb{Q}$ is S_5 .

Exercise 3 Do Problems 10, 11 and 12 in Exercises 57.13, pp.349–351.